

ISKANDAR KHOLMANOV

DATA SCIENTIST/ML ENGINEER

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Green card

U.S. Work Authorization – No visa sponsorship required (now or in the future)

SUMMARY

Data Scientist/ML **Engineer** with a PhD in Physics and 8+ years solving complex, data-intensive problems across academic research and industry. Experienced in Python and SQL, and in designing, building, and deploying **cloud-native ML and GenAI systems** with API-based architectures, scalable data pipelines, and production monitoring. Hands-on experience with **LLM fine-tuning, retrieval-augmented generation (RAG), and end-to-end model deployment**, with a strong focus on reliability, performance, and cost-aware system design. Co-author of 40+ peer-reviewed publications and a U.S. patent.

SKILLS

- ❖ **Programming & MLOps:** Python, SQL, Git, Docker, CI/CD (GitHub Actions), Databricks (Spark), AWS (EC2, S3, Lambda, API Gateway), Serverless ML, model serving, CloudWatch (metrics, logs, dashboards), latency & cost monitoring, MLflow, Airflow, cloud-native deployment;
- ❖ **GenAI & LLMs:** LLM fine-tuning (LoRA/PEFT), end-to-end RAG pipelines, LangChain, LlamaIndex, Hugging Face, VectorDB (FAISS, ChromaDB), semantic search, LLM inference APIs (Groq, LLaMA-3.x);
- ❖ **ML & DL:** PyTorch, TensorFlow, scikit-learn, classical ML, deep learning, transformer architectures;
- ❖ **Statistical Analysis:** EDA, A/B testing, RAG evaluation analysis, time series analysis, statmodels;
- ❖ **Leadership & Operations:** Cross-functional collaboration, data-driven decision-making, production ML ownership, operational optimization, cost-aware system design.

EXPERIENCE

- ❖ **Data Scientist / Machine Learning Engineer** | TripleTen | 2024–Present
 - Building and deploying GenAI systems using RAG workflows, S3, FastAPI, Implementing cloud observability;
 - Built **LLM fine-tuning pipelines** (qLoRA/PEFT) and inference workflows optimized for CPU/GPU environments;
 - Led end-to-end ML models (linear, tree-based and boosting) including deployment, monitoring and validation;
- ❖ **Founder & Business owner** | Uzeat LLC | 2017-2024
 - Provided decisive COVID-19 crisis leadership, preserving margins through forecasting and cost control.;
 - Led strategic planning to drive revenue growth and profitability across key business units.
- ❖ **Research Scientist**, The University of Texas at Austin | 2010 – 2017
 - Led data-driven research resulting in 40+ papers, a U.S. patent, and industry partnerships. | [Github](#)

TECH PROJECTS

- ❖ **Production RAG System with Cloud Observability & Cost-Aware LLM Serving.** Built an end-to-end RAG pipeline using **sentence-transformers (MiniLM)**, **FAISS-based retrieval**, and **Groq-hosted LLaMA-3.1-8B** for generation; implemented decoupled **Colab→AWS** architecture (S3 + EC2) with **FastAPI** serving layer and **CloudWatch**-based observability including structured logging, latency decomposition (embedding/FAISS/LLM), token usage tracking, and p95 latency analysis; generated 60+ request tests to validate system stability and production-level monitoring. | [Github](#)
- ❖ Built an **end-to-end LLM fine-tuning pipeline (Mistral-7B + LoRA/PEFT)** using **Google Colab (T4 GPU)** for efficient training and experimentation. Created instruction-tuning datasets (system/user/assistant format) using Amazon reviews, synthetic Q&A, and product metadata. Deployed a **4-bit quantized inference pipeline (QLoRA)** with **FastAPI**, exposed via **ngrok for real-time chat**. Analyzed model behavior and identified dataset-scale and instruction-alignment limits affecting LoRA performance. | [Github](#)
- ❖ **Stock Price Forecasting.** Built a production-grade time-series ML system using **PyTorch LSTM**, orchestrated with **Dockerized Airflow**, tracked via **MLflow**, monitored with **Evidently**, and **deployed on Streamlit Cloud** for live inference; ingested Yahoo Finance data; benchmarked against **SARIMA, Random Forest, XGBoost, and LSTM-Keras** models. The PyTorch LSTM achieved **1.64% MAPE** on unseen data, outperforming all baselines | [Github](#) | [Streamlit App](#)

EDUCATION

- **TripleTen Bootcamp** | Data Science / Machine Learning Certificate | 2024;
- **PhD in Physics**, Academy of Science of Uzbekistan;